

Problem Statement: Automating Competitive Price Intelligence to Protect Retail Profit Margins

1. Background

In highly volatile e-commerce and retail market sectors, tracking competitor pricing is a critical business function. For a scaling retailer, keeping prices synchronized with the market directly impacts sales volume and gross margin health. However, manual tracking across dozens of competitor websites introduces massive operational overhead and severe data collection delays.

2. The Core Challenge

Relying on manual human workflows to monitor market pricing introduces critical business vulnerabilities:

- **Operational Inefficiency:** Manually auditing web catalog prices requires employees to spend several hours a week on repetitive copy-paste workflows, inflating labor costs.
- **Human Error Risk:** Manual transcriptions are prone to data entry errors, which can result in incorrect pricing updates and lost revenue.
- **No Real-Time Insights:** Prices change dynamically. By the time a manual audit spreadsheet is completed, the data is stale, preventing the business from taking agile action.

3. Objective & Overarching Business Question

"How can we build an automated data acquisition engine that replaces manual data retrieval workflows, calculates real-time price deltas against market competitors, and quantifies structural business hours saved?"

4. Project Deliverables

To deliver clear strategic value to business stakeholders, this project builds:

1. **Automated Ingestion Layer (Python Requests):** An efficient network connector that dials into competitor web infrastructures and grabs real-time catalog trees.
2. **Parsing & Extraction Engine (BeautifulSoup4):** A structural extraction layer optimized to isolate item descriptions, SKU labels, and cost metrics from raw HTML layouts.
3. **Price Intelligence Logic:** Functional business calculations that evaluate "Price Delta" metrics to identify over-pricing or under-pricing gaps.
4. **Data Delivery Output:** Structured exports saved directly to downstream local environments for immediate analysis or business intelligence ingestion.